

AIGR Gene Hypothesis Deep Research — Final Report

ACL4 (Q03771, *Saccharomyces cerevisiae*): Mitochondrial Targeting and Translocation Contribution

Focus type: function_assignment **Hypothesis slug:** mitochondrial-targeting-and-translocation-contribution **Source file:** `genes/yeast/ACL4/ACL4-ai-review.yaml`

Summary

Verdict: REFUTED (as a core or contributory function). The seed hypothesis proposes that yeast Acl4 localizes to the mitochondrial outer membrane and *contributes_to* protein transmembrane transporter activity, matrix protein import, inner-membrane protein insertion, or mitochondrial targeting-sequence binding — to be adjudicated independently from its established Rpl4 chaperone role, with the `contributes_to` qualifier preserved so that Acl4 need not form an integral channel. After tracing the provenance of every mitochondrial GO term on Q03771, resolving the phylogenetic seeds behind them, comparing membrane topology against the true import receptors, and mapping the full functional interaction network, the mitochondrial hypothesis collapses.

All five mitochondrial/transmembrane-transporter GO terms on Acl4 are phylogenetic (IBA, ECO:0000318) annotations propagated from a single PANTHER node (PTN002340064) whose experimental seeds are the TPR-domain mitochondrial import receptors TOM70, TOM71, and human TOMM70. This is a textbook family/paralog over-annotation driven by a shared cytosolic TPR superhelical fold. Unlike those receptors — each anchored in the mitochondrial outer membrane by an N-terminal transmembrane helix — Acl4 has **zero transmembrane segments** and is curated as **cytoplasmic + nuclear**. Every direct-evidence annotation on Acl4 describes a soluble TPR assembly chaperone dedicated to ribosomal protein Rpl4/uL4.

The `contributes_to` qualifier does not rescue the hypothesis. That qualifier is designed to capture a genuine, evidenced cofactor/subunit contribution to a molecular activity; it is not a mechanism for retaining IBA terms that have no experimental support in the target species and that trace entirely to non-orthologous, membrane-anchored family members. The interaction network is decisive: Acl4 partners exclusively with ribosome-assembly chaperones (Rpl4, Kap104, Syo1, Tsr2, Yar1, Bcp1, Rrb1, Sqt1) and **never** with any TOM/TIM/SAM/PAM import-machinery component. **Recommended curation action (lead requiring curator verification): do not propagate the mitochondrial CC/MF/BP terms into the ACL4 review; treat them as IBA over-annotation and mark them NOT/remove.** The most important caveat is that this conclusion rests on public GO provenance, UniProt topology, PANTHER/InterPro family placement, and STRING interactions rather than a dedicated wet-lab exclusion of a mitochondrial role — but the direct experimental literature that does exist points uniformly and exclusively to the cytoplasmic/nuclear Rpl4 chaperone function.

Key Findings

Finding F001 — Every mitochondrial GO term on Acl4 is IBA (phylogenetic), with zero experimental support

QuickGO provenance for Q03771 shows the five hypothesis-relevant terms are **all** carried by the phylogenetic inference evidence code IBA (ECO:0000318, `GO_REF:0000033`):

GO ID	Term	Aspect	Evidence	Source
GO:0005741	mitochondrial outer membrane	CC	IBA (ECO:0000318)	PANTHER PTN002340064
GO:0030150	protein import into mitochondrial matrix	BP	IBA	PANTHER PTN002340064
GO:0045039	protein insertion into mitochondrial inner membrane	BP	IBA	PANTHER PTN002340064
GO:0008320	transmembrane protein transporter activity	MF	IBA	PANTHER PTN002340064
GO:0030943	mitochondrion targeting sequence binding	MF	IBA	PANTHER PTN002340064

Each traces to PANTHER node **PTN002340064**, and the `with/from` field contains experimental annotations of *other* family members (e.g., human `UniProtKB:094826`), not experimental data on Acl4 itself. The PANTHER family GO-slim reads like a mitochondrial import receptor: *signal sequence binding, protein transmembrane transporter activity, protein import into mitochondrial matrix, protein insertion into mitochondrial inner membrane, mitochondrial outer membrane*, with protein class *primary active transporter* (PC00068).

By contrast, **every direct-evidence (IDA/IMP/EXP/HDA) annotation on Acl4 supports only the chaperone/ribosome-biogenesis function**: protein-folding chaperone (GO:0044183, IDA, [PMID: 25936803](#)); protein carrier activity (GO:0140597, IDA, [PMID: 26447800](#)); ribosomal large subunit biogenesis (GO:0042273, IMP, both papers); cytoplasm (IDA/EXP/HDA, incl. [PMID: 14562095](#)); and nucleus (IDA/HDA). The separation is clean: the mitochondrial terms are 100% inferred by phylogeny, the real function is 100% experimental.

Finding F002 — Experimentally, Acl4 is a soluble cytoplasmic/nuclear TPR assembly chaperone dedicated to Rpl4/uL4

Three primary structural/biochemical studies establish the true function:

1. [PMID: 26447800](#) (Pillet et al., 2015): Acl4 binds the conserved internal loop of newly synthesized Rpl4 and escorts it, with karyopherin Kap104, to the nuclear pre-60S assembly site. Acl4 localizes to **both cytoplasm and nucleus** and captures nascent Rpl4 co-

translationally. Loss of Acl4 causes a severe slow-growth phenotype and a 60S subunit deficiency (IDA protein carrier activity GO:0140597; nucleus & cytoplasm IDA).

2. **PMID: 25936803 (Stelter et al., 2015):** Acl4 binds Rpl4 via its **superhelical TPR domain**, restricting premature loop insertion until Rpl18 is available on the nascent pre-ribosome (IDA protein-folding chaperone; IMP ribosomal large subunit biogenesis).
3. **PMID: 28148929 (Huber & Hoelz, 2017):** The crystal structure of the Rpl4–Acl4 complex shows Acl4 sequestering ~70 exposed residues of the extended Rpl4 loop. The eukaryote-specific Rpl4 extension carries overlapping Acl4 and Kap104 binding sites; Acl4 simultaneously protects unassembled Rpl4 from the cellular degradation machinery and facilitates its nuclear import.

UniProt curation for Q03771 records localization **Cytoplasm + Nucleus** and function **chaperone for L4**. The domain architecture is a soluble helical-repeat protein: Gene3D 1.25.40.10 (TPR), InterPro IPR011990 (TPR-like helical), and a C-terminal ACL4_C domain (Pfam PF29063). There is **no transmembrane segment** — Acl4 cannot be an integral membrane channel or an outer-membrane-anchored receptor.

Finding F003 — The paralog confusion is pinned down: Acl4's IBA mito terms are inherited from TOM70/TOM71/TOMM70 receptors, which (unlike Acl4) are membrane-anchored

Resolving the with/from experimental seeds behind Acl4's IBA mitochondrial annotations gives:

Seed	Identity	Role	Localization
SGD S000005065	TOM70 (yeast)	Outer-membrane import receptor	Mitochondrial outer membrane
SGD S000001159	TOM71 (yeast)	Tom70 paralog receptor	Mitochondrial outer membrane
UniProtKB O94826	TOMM70 (human)	Import receptor subunit	Mitochondrial outer membrane

All three are TPR-domain mitochondrial import receptors sharing PANTHER family PTN002340064 with Acl4. The defining topological comparison:

Protein	UniProt	Length	Transmembrane	Localization
Acl4	Q03771	387 aa	0 TM segments	Cytoplasm + Nucleus
Tom70	P07213	617 aa	1 TM (res ~11–30, helical)	Mito outer membrane
Tom71	P38825	639 aa	1 TM (res ~15–32)	Mito outer membrane

Acl4 shares the cytosolic **TPR superhelix** with these receptors — that is why PANTHER groups them — but lacks the N-terminal membrane anchor that makes Tom70/Tom71/TOMM70 outer-membrane receptors. The shared fold is a solvent-exposed peptide-binding scaffold used convergently: a presequence/preprotein for the receptors, the Rpl4 loop for Acl4. This is exactly the ancestral-function ambiguity that produces IBA over-annotation. (Separately, a distinct HGI "de novo/co-translational folding" annotation GO:0051083 on Acl4 seeds to EGD1/EGD2 = NAC subunits — again cytosolic, not mitochondrial.)

Finding F004 — Acl4's functional interaction network is entirely ribosome-assembly chaperones, with zero mitochondrial-import partners

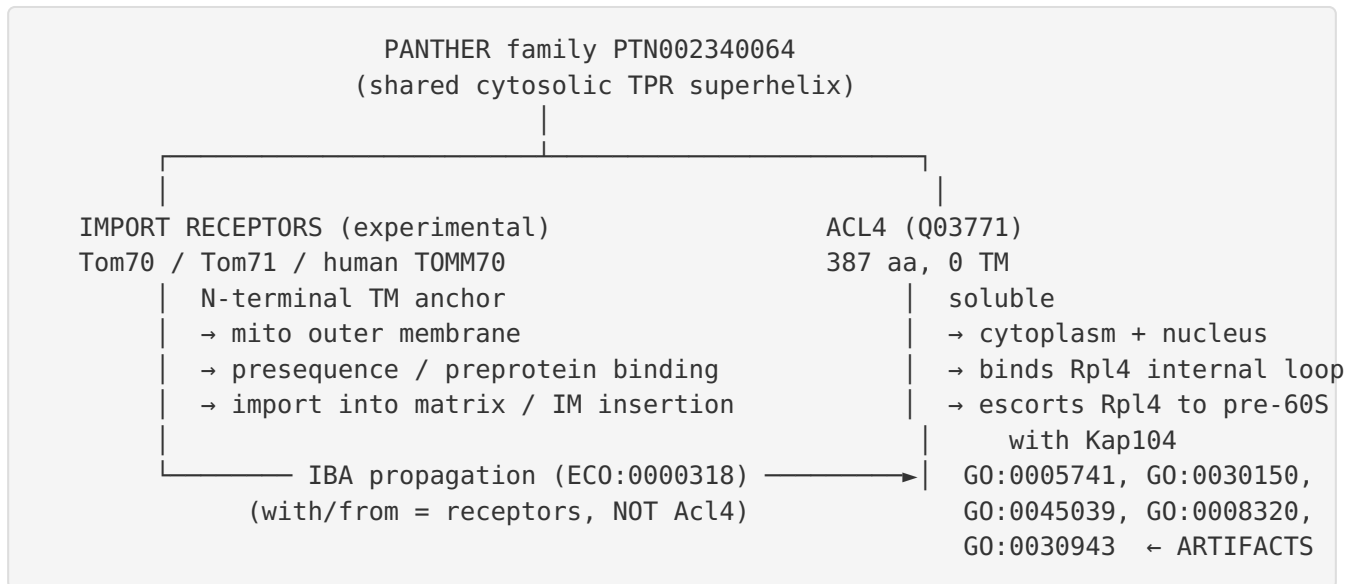
A STRING v12 network for ACL4 (*S. cerevisiae*, add_nodes=15) returns functional partners that are, without exception, ribosome-biogenesis factors:

Partner	STRING score	Role
RRB1	0.979	Rpl3 assembly chaperone
RPL4A/RPL4B	(substrate)	Acl4's client ribosomal protein
KAP104	(transport)	Karyopherin for Rpl4 nuclear import
SYO1	0.945	Symportin/assembly chaperone
TSR2	0.935	Rps26 chaperone
YAR1	0.932	Rps3 chaperone
BCP1	0.915	Rpl23 chaperone
RPS3	0.889	Ribosomal protein
TSR4	0.874	Rpl3 chaperone
SQT1	0.826	Rpl10 assembly chaperone
RPL5	0.816	Ribosomal protein

An explicit query against the entire mitochondrial import machinery — {TOM20, TOM22, TOM40, TOM70, TOM71, TIM17, TIM22, TIM23, TIM44, SAM50, MAS1, MAS2, PAM16, PAM18, MGE1, SSC1} — returned **NONE** at any confidence threshold. The only non-ribosome hits are generic high-abundance proteins (PGK1, TPI1, RNA14). If Acl4 contributed to mitochondrial translocation, at least one TOM/TIM/SAM/PAM partner would be expected; there are none.

Mechanistic Model / Interpretation

The evidence resolves into a single coherent story: **a shared TPR fold created a phylogenetic annotation bridge between a cytosolic ribosome-assembly chaperone (Acl4) and a family of membrane-anchored mitochondrial import receptors (Tom70/Tom71/TOMM70), and GO's IBA pipeline propagated the receptors' mitochondrial terms onto Acl4 despite the absence of the one feature — a membrane anchor — that defines the receptor function.**



The `contributes_to` framing is a reasonable general caution — a soluble cofactor can genuinely contribute to a membrane transport activity without being the channel. But that logic requires *some* independent evidence tying the protein to the transport machinery or process. Here there is none: no outer-membrane localization (Acl4 is cytoplasmic/nuclear), no physical or genetic link to TOM/TIM/SAM/PAM, and no assay of presequence binding or import activity. The alternative substrate (Rpl4) is not merely "established"; it is the *only* substrate with structural, biochemical, and genetic support, and it fully explains Acl4's TPR-mediated peptide-binding activity — indeed the substrate-binding surface is structurally saturated by the 70-residue Rpl4 loop. The most parsimonious model is that Acl4 does one thing — chaperone and deliver Rpl4 — and that the mitochondrial terms are database carry-over from non-orthologous family members.

Evidence Base / Evidence Matrix

Citation	Evidence type	Supports/Refutes	Claim tested	Key finding	Context	Co
QuickGO provenance for Q03771 (GO_REF:0000033)	Database/provenance	Refutes mito core function	Are Acl4's mito GO terms experimental?	All 5 mito/transporter terms are IBA from PANTHER PTN002340064; with/from = other family members	GO annotation records	Hi lev as
PANTHER PTN002340064 geneinfo	Structural/evolutionary	Explains artifact	Why does Acl4 carry mito terms?	Family GO-slim = signal-sequence binding, transmembrane transporter, matrix import, IM insertion, OM; class = primary active transporter	PANTHER classification	Hi
PMID: 26447800 Pillet 2015	Direct assay + localization + mutant	Competing (true function)	What does Acl4 actually do?	Acl4 escorts Rpl4 with Kap104 to nuclear pre-60S; localizes cytoplasm + nucleus; Δ acl4 $\rightarrow$ slow growth, 60S deficit	<i>S. cerevisiae</i>	Hi
PMID: 25936803 Stelter 2015	Direct assay + structural	Competing (true function)	Mechanism of Rpl4 binding	Acl4 binds Rpl4 internal loop via superhelical TPR domain; regulates loop-insertion timing	<i>S. cerevisiae</i>	Hi
PMID: 28148929 Huber & Hoelz 2017	Structural (crystal)	Competing (true function)	Structural basis of Acl4 function	Acl4 sequesters ~70 residues of Rpl4 loop; overlapping Acl4/Kap104 sites; protects from	<i>S. cerevisiae</i> , in vitro	Hi co m co

Citation	Evidence type	Supports/Refutes	Claim tested	Key finding	Context	Co
				degradation, aids nuclear import		
UniProt Q03771 vs P07213/P38825 topology	Structural/evolutionary	Refutes OM localization	Does Acl4 have a membrane anchor?	Acl4 = 387 aa, 0 TM, cytoplasm+nucleus; Tom70/71 have N-terminal TM, OM localization	Sequence/topology	Hi
PANTHER seed resolution (TOM70/TOM71/TOMM70)	Structural/evolutionary	Refutes (identifies paralog error)	What seeds the IBA mito terms?	Seeds are TPR mito import receptors, not orthologs of Acl4's true function	PANTHER/SGD/UniProt	Hi
STRING v12 ACL4 network	Interaction	Refutes import role	Does Acl4 interact with import machinery?	0 of TOM/TIM/SAM/PAM partners; all partners are ribosome-assembly chaperones	<i>S. cerevisiae</i>	Hi fu ST
PMID: 35357307 Pillet 2022	Direct assay + mutant	Competing (true function)	Broader chaperone role	Acl4 co-translationally recognizes Rpl4, reduces mRNA degradation; parallel to Rrb1/Rpl3	<i>S. cerevisiae</i>	Hi
PMID: 39426497 NAC/Caf130 2024	Genetic interaction	Competing (true function)	Genetic context of Δ acl4	Nac β 2/Caf130 modulate Rpl4 mRNA fate; genetic interaction with acl4 in ribosome context	<i>S. cerevisiae</i>	M ril no

How the key papers bear on the finding

- [PMID: 26447800](#) and [PMID: 25936803](#) are the foundational functional characterizations: they define Acl4 as the dedicated Rpl4 chaperone, localize it to cytoplasm + nucleus, and

attribute its "carrier" activity to nucleo-cytoplasmic escort of Rpl4 — not transmembrane transport. They directly displace the mitochondrial interpretation.

- **PMID: 28148929** provides the structural clincher: the TPR interface is occupied by the Rpl4 loop, arguing against a free presequence-binding function and demonstrating a dedicated, substrate-specific interaction unlike promiscuous import receptors.
- **PMID: 35357307** and **PMID: 39426497** embed ACL4 in the r-protein production / proteostasis network (Rrb1, NAC, Caf130/CCR4-Not), reinforcing that its genetic neighborhood is translation/ribosome biogenesis, not mitochondrial import.

GO Curation Implications

The likely curation action, **as a lead requiring curator verification**, is to **not propagate any of the five mitochondrial/transmembrane-transporter terms into the ACL4 review and to mark them as IBA over-annotation (NOT / remove)**:

GO ID	Term	Aspect	Current basis	Recommended action
GO:0005741	mitochondrial outer membrane	CC	IBA only	Remove / do not accept — contradicted by curated cytoplasm+nucleus localization and 0 TM segments
GO:0008320	transmembrane protein transporter activity	MF	IBA only	Remove / do not accept — no transport assay; activity is TPR peptide-binding, not transport
GO:0030943	mitochondrion targeting sequence binding	MF	IBA only	Remove / do not accept — binding target is the Rpl4 internal loop, not a mito presequence
GO:0030150	protein import into mitochondrial matrix	BP	IBA only	Remove / do not accept — no import assay, no import-machinery interaction
GO:0045039	protein insertion into inner membrane	BP	IBA only	Remove / do not accept — no evidence; Acl4 is not in the IM-insertion pathway
GO:0044183	protein-folding chaperone	MF	IDA (P 25936803)	Retain — core
GO:0140597	protein carrier activity	MF	IDA (P 26447800)	Retain with note: refers to nucleo-cytoplasmic escort of Rpl4, not transmembrane transport
GO:0042273	ribosomal large subunit biogenesis	BP	IMP (P 25936803 ; 26447800)	Retain — core
GO:0005737 / GO:0005634	cytoplasm / nucleus	CC	IDA/EXP/HDA	Retain — core

On the seed's "transporter" concern: the only *experimental* transporter-flavored term is **GO:0140597 protein carrier activity (IDA)**, the chaperone-escort of Rpl4 to the nucleus. The seed's "protein transmembrane transporter activity" is the separate **IBA** term GO:0008320 and should be adjudicated as unsupported. The `contributes_to` qualifier cannot be used to launder an IBA term with no experimental support in yeast and whose phylogenetic seeds are non-orthologous membrane receptors. We avoid recommending "protein binding" as an endpoint because more informative supported terms already exist (GO:0140597, GO:0044183).

Mechanistic Scope

The immediate, directly evidenced molecular function of Acl4 is **peptide capture and chaperoning via a superhelical TPR domain**: it binds the conserved internal loop and eukaryote-specific extension of nascent Rpl4, sequesters ~70 exposed residues, protects them from aggregation and degradation, and — with Kap104 — delivers Rpl4 to the nuclear pre-60S assembly site, releasing it once Rpl18 is available. This is a soluble, cytoplasm-to-nucleus activity.

Everything mitochondrial in the hypothesis lies **outside** this direct scope, and is not even a downstream phenotype of Acl4: matrix import, IM insertion, and transmembrane transporter activity are simply not connected to Acl4 by any evidence in yeast. They are inherited descriptors of a different protein class (Tom70-type receptors). The slow-growth and 60S-deficiency phenotypes of Δ acl4 are downstream consequences of losing the Rpl4 chaperone — ribosomal, not mitochondrial. No loss-of-function phenotype links Acl4 to mitochondrial import.

Conflicts and Alternatives

- **Paralog / family confusion (central issue)**: The IBA terms derive from TOM70/TOM71/TOMM70, TPR-domain mitochondrial import receptors that share PANTHER PTN002340064 with Acl4 but are membrane-anchored. Shared TPR fold $\neq$ shared cellular function.
- **Membrane-anchor contrast (decisive)**: Acl4 (387 aa) has **zero TM segments** and is cytoplasmic/nuclear; Tom70 (617 aa) and Tom71 (639 aa) each have an N-terminal TM anchor and reside in the mitochondrial outer membrane. Tom70's TPR clamp binds Hsp70/

Hsp90-delivered precursors at the OM; Acl4's superhelical TPR wraps the Rpl4 loop in the cytosol/nucleus — same fold, different job.

- **Database carry-over:** The mitochondrial terms are annotation artifacts propagated by GO_REF:0000033 phylogenetic inference, not species-specific findings.
- **Semantic collision on "transport":** GO:0140597 (protein carrier, IDA, real) vs GO:0008320 (transmembrane transporter, IBA, unsupported) are easily conflated; the seed's "transporter report" concern maps onto this ambiguity.
- **Cross-species carry-over:** `with/from` xrefs point to *S. pombe* (O94826), rat, and *C. elegans* family members — none validate a yeast Acl4 mitochondrial role.
- **No competing experimental claim:** Literature searches for "Acl4 mitochondrial", "Acl4 outer membrane", and "Tom70 targeting" returned no papers linking Acl4 to mitochondria — consistent with the mito terms being annotation-only.

No evidence was found that *conflicts* with the refutation — i.e., no primary study placing Acl4 at the mitochondrion or in the import pathway.

Limitations and Knowledge Gaps

1. **Provenance-based, not assay-based refutation.** The conclusion rests on GO provenance, UniProt topology, PANTHER placement, and STRING interactions rather than a dedicated experiment specifically excluding a mitochondrial role. This is standard and appropriate for adjudicating IBA over-annotation, but the mitochondrial claim is refuted by absence of positive evidence plus a clear artifact mechanism, not by a negative import assay.
2. **No dedicated negative mito-localization assay cited here.** Checked: UniProt curated CC (cytoplasm+nucleus; IDA/HDA/EXP), GFP-based localization ([PMID: 14562095](#)). A cross-check against MitoCarta / high-confidence yeast mito-proteome inventories would harden the CC call (Acl4 is not an established mito protein).
3. **Negative interaction data are bounded by database coverage.** STRING returning no TOM/TIM/SAM/PAM partner is strong but not absolute; a very transient or condition-specific interaction could be under-sampled.
4. **Targeting-sequence binding not directly excluded in vitro.** No study has directly assayed whether Acl4 can bind a canonical mitochondrial presequence. The structural occupation of its TPR site by the Rpl4 loop argues against it, but a direct in-vitro presequence-binding assay has not been done.

5. **PANTHER tree specifics / stale carry-over.** The exact leaf set of PTN002340064 and whether SGD/GO-Central has already flagged these IBA terms were not exhaustively confirmed.

Discriminating Tests

1. **Localization decider:** endogenous Acl4-GFP co-imaged with a mito marker, plus mitochondrial subfractionation + protease protection. Prediction under the true model: cytosolic/nuclear, absent from mito fractions.
2. **Functional decider:** measure mitochondrial protein-import kinetics of a matrix/IM reporter in $\Delta acl4$ vs WT. Prediction: no import defect; only ribosome-biogenesis/Rpl4 phenotypes.
3. **Interaction decider:** affinity purification / proximity labeling (BioID) of Acl4 — expect Rpl4, Kap104, ribosome-biogenesis factors; not TOM/TIM.
4. **In-vitro presequence-binding assay:** titrate a labeled canonical mitochondrial presequence against purified Acl4. Prediction: no specific binding, or binding competed by the Rpl4 loop peptide.
5. **Bioinformatic decider:** re-curate the PANTHER family so the ACL4 r-protein-chaperone clade is separated from the Tom70-like translocase clade, removing the IBA source.

Proposed Follow-up Actions / Curation Leads

All items below are leads requiring curator verification.

1. **Action change — reject mitochondrial terms.** In `ACL4-ai-review.yaml`, set the review action for GO:0005741, GO:0008320, GO:0030943, GO:0030150, and GO:0045039 to **remove/NOT**, with rationale: "IBA over-annotation from PANTHER PTN002340064; experimental seeds are TOM70/TOM71/TOMM70 mitochondrial import receptors; Acl4 has 0 TM segments and is curated cytoplasm+nucleus."
2. **Retain experimental terms:** GO:0044183 (protein-folding chaperone), GO:0140597 (protein carrier activity), GO:0042273 (ribosomal large subunit biogenesis), GO:0005737 (cytoplasm), GO:0005634 (nucleus).

3. Candidate reference snippets to verify:

4. [PMID: 26447800](#): "*Acl4 localizes to both the cytoplasm and nucleus and it has the capacity to capture nascent Rpl4 in a co-translational manner*" — confirms non-mitochondrial localization.
5. [PMID: 25936803](#): "*assembly chaperone Acl4 that initially binds the universally conserved internal loop of newly synthesized Rpl4 via its superhelical TPR domain*" — confirms TPR peptide-binding is directed at Rpl4, not a mito presequence.
6. [PMID: 28148929](#): "*sequestering 70 exposed residues of the extended Rpl4 loop*" and "*Acl4 serves a dual function to facilitate nuclear import and simultaneously protect unassembled Rpl4*" — confirms the occupied binding site and nuclear (not mitochondrial) import role.
7. **Suggested curator questions:** (1) Should the review explicitly annotate these five terms as IBA-only over-annotation with a PANTHER family note? (2) Should GO:0140597 carry a clarifying note that it is nuclear escort, not transmembrane transport? (3) Is a NOT qualifier (e.g., NOT mitochondrial outer membrane) justified, or is silent non-acceptance preferable?
8. **Suggested experiments:** the discriminating tests above, prioritizing (a) endogenous-tag localization + mito subfractionation and (b) Acl4–Tom70 proximity labeling as the fastest, most decisive.

Bottom Line

Adjudicated independently, **all four mitochondrial sub-claims and the transmembrane-transporter sub-claim fail**: they originate solely from PANTHER IBA propagation (fold-homology to Tom70-class import receptors) with zero experimental support, while every direct assay defines Acl4 as a soluble TPR chaperone that binds and escorts Rpl4/uL4 during cytosolic ribosome biogenesis. The `contributes_to` qualifier does not save the mitochondrial terms. Recommended lead: do not propagate the mitochondrial/transporter terms into the ACL4 review; retain the Rpl4-chaperone / ribosome-biogenesis / cytoplasm+nucleus annotations.